

Answers to Referees' Questions

“Interpolating the Stochastic Heat and Wave Equations with Time-independent Noise: Solvability and Exact Asymptotics”

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We would like to thank the two anonymous reviewers for carefully reading our manuscript and for providing detailed reports, which have helped us to improve significantly the paper. We have revised our paper accordingly. Here are a list of changes that we have made. The page and equation numbers refer to the revised version of the manuscript.

The page and line numbers below refer to those from the original submission. In particular, the line number refers to the one added by the system at the left margin of the paper.

1 Changes according to Reviewer 1

1. page 2: Fixed the “nonnegative and nonnegative-definite” typo.
2. page 4: **I feel that adding too much info will disrupt the flow of the introduction. I added that $\mathcal{M}$ is finite which is really all that is all that matters.**
3. page 13: Fixed the typo involving $\mathcal{A}_{a,d}$.
4. page 15: We added a reference that thoroughly treats the processes of defining the Wiener Chaos expansion of the solution and proving its validity.
5. page 21: **(Did you update this one? Please double check)** All $\sigma(d, \alpha)$ in the proof of Lemmas 3.10 and 3.11 were replaced with $\sigma(a, d, \alpha)$.
6. page 29: “Now are are” has been updated to “Now we are”. **(You forgot this one.)**

2 Changes according to Reviewer 2

1. page 1, line 21, 23 and 24: Changed “exits” to “exists”.

2. page 1, line 22: Inserted “as that” to the sentence.
3. page 2, line 32: Added parenthesis to $f^{(m)}$.
4. page 2: We removed the mention of Z and Z^* and then rewrote the last paragraph to accommodate this change.
5. page 2, line 53: The homogeneous solution is the constant 1. The stochastic integral takes into consideration the driving force. Then by superposition, we add the homogeneous solution with the stochastic integral.
6. page 4, line 48: Corrected the “Wiener” misspelling.
7. page 5: We corrected the condition on r so that $r < 1/2$.
8. page 5, line 15: We added a space after “ $p \geq 2, \dots$ ”.
9. page 6, line 23: Added the word “illustrate”.
10. page 6, line 34: Changed “exits” to “exists”.
11. page 7, item (b): We referenced the theorem and figure below.
12. page 7, line 7: Changed “to” to “with”.
13. page 7, line 14: We removed the word “the”.
14. page 7, line 37 and line 48: Changed “exits” to “exists”.
15. page 7, line 41: Rewrote the paragraph starting on this line.
16. page 10, line 44: Changed “Laplace” to “Laplacian”.
17. page 10: We removed the repeated $\alpha = a = d = 2$ case from the bottom of the page.
18. page 13, line 27: Added that $\alpha < a$ as well.
19. page 13: Fixed the $\gamma(x - y)$ typo.
20. page 14: Example 2.10 refers to Example 2.4 and in that example, other papers are cited so it seems that this example should stay. Example 2.11 needs to refer to Example 2.5 (there was a typo that I fixed) and again Example 2.5 refers to other papers.
21. page 14, line 3: Changed “exits” to “exists”.
22. page 15, line 11: Added a reference to the Skorohod integral at the end of this paragraph.
23. page 15, line 18: I believe that the reader will now be clear about the definition of G since we have added additional information about G on page 2.

24. page 15, line 31: Reworded the sentence to say that the moment “diverges to infinity” for $t > T_2$.
25. page 15, line 39: Fixed the Wiener misspelling.
26. page 16, line 18: I believe that the reader will now be clear about the definition of G since we have added additional information about G on page 2.
27. page 16, line 20: Fixed the order of the arguments for the kernel $\tilde{f}_n$.
28. page 16, line 38: Fixed the “Wiener” misspelling.
29. page 16, line 40: Changed R to $\mathbb{R}$.
30. page 17, line 23: Added the assumption that $\alpha < \min\{2a, d\}$.
31. page 17, Theorem 3.5: Added the assumption that $\alpha < \min\{2a, d\}$.
32. page 20: Replaced “ f ” with “ b ” in the definition for $\|\cdot\|_{q,w}$.
33. page 20, line 23: Removed the reference to [1, Lemma 1.6].
34. page 21, line 9: Added an equation link to where this is referenced.
35. page 21, Lemma 3.9: Added the assumption on α . Directed the reader to the definition of $\sigma(a, d, \alpha)$. Replaced “2” with “ α ” in the exponent.
36. page 21, Lemma 3.10: Made clear that Lemma 3.5 is from the current paper. Fixed the g typo inside the definition of Λ .
37. page 22, line 23: Replaced “ $\mathbb{R}$ ” with “ $\mathbb{R}_+$ ”.
38. page 22, line 33: Replaced “2” with “ α ” in the exponent.
39. page 22: All $\sigma(d, \alpha)$ were replaced with $\sigma(a, d, \alpha)$.
40. page 22: We reverted back to using γ instead of the Riesz kernel. ie equation (3.34).
41. page 22, Lemma 3.11: Replaced $\gamma(x - y)$ with $\gamma(x + y)$.
42. page 23, line 4: Corrected the centered equation.
43. page 23: Reverted back to using $\gamma(\cdot)$ instead of $|\cdot|^{-\alpha}$.
44. page 23, line 48: Replaced “ f ” with “ g ”.
45. page 24, line 3: Added reference to Lemma 3.11 for the use of σ .
46. page 24: Removed the excess periods, fixed the exponents by changing “2” to “ α ”, and also replaced “ $\sigma(d, \alpha)$ ” with “ $\sigma(a, d, \alpha)$ ”.
47. page 25, line 3: Made it clear we are only proving part I.

48. page 25, Eq 4.5: Fixed the $\xi_{\sigma(j)}$ error.
49. page 25, line 50: Gave a reference to the function K .
50. page 26, Eq 4.6: Explained the notation.
51. page 26 and 27, Eq 4.2: $\alpha < 2a$ was assumed in the statement of the theorem and we added a reference to clarify that. We also reordered the argument as suggested.
52. page 27, Eq 4.14: Replaced “ n ” with “2”.
53. page 27, line 28: Added a reference for the inequality.
54. page 27, line 35: Made a comment about the switch to $x = 0$.
55. page 28, Lemma 5.1: Relaxed the assumption on α and referenced ρ as was suggested.
56. page 28, line 59: Added a reference for H .
57. page 29: Fixed all the +1 typos.
58. page 29: Explained the derivation of C_n .
59. page 29, line 56: Fixed the misspelling of hypercontractivity.
60. page 30: You were correct, we made the change to r ie $r \in [0, 1/2)$.
61. page 30, line 32: Fixed the typo involving the reference to Lemma 5.1.
62. page 32, line 37: Added “it”.
63. page 33, line 22: Fixed the subscript in $\mathbb{E}(U_n(t))$.
64. page 33, line 50: Replaced H with $\mathcal{H}$.
65. page 34, line 46 Unsure what to change and the the $\rho(g)$ comment.
66. page 34, line 46: Added that $\mathcal{F}^{-1}g \in \mathcal{H}_+$ which in return implies that $g \in L^2(\mu(dx), \mathbb{R}^d)$
67. page 34, line 49: We have altered the argument to bound from below the quantity $\rho(g)$ by they same thing but with the sup only over nonnegative h . In addition, we replaced $\rho(g)$ with $\rho'(g)$ so there is no confusion with the previous ρ defined in the paper.
68. page 35, line 13: Unsure what to say about the comment on $\mathcal{F}g_h * \widetilde{\mathcal{F}g_h}$.
69. page 35: Fixed all g_h typos.
70. page 37, Eq 6.19: Fixed the n_t typo.

- 71. page 38, line 10: Fixed the t_p typo.
- 72. page 39: Reformatted the proof of (4.9) and took into account the suggestions made.
- 73. page 40: Removed the duplicated reference.

3 Additional minor corrections

In addition to the above changes suggested by the two referees, we implemented some minor corrections which are highlighted in blue in the manuscript. We list these below.

- 1. page 31, Lemma 5.2: In the proof starting with the line “We want to find β and A so that...” we replaced R_n with R^n in the following centered equation.
- 2. Page 35, Prop 6.4: In the proof, we removed the line that calculated $\|g\|_{W^{1,a}(\mathbb{R}^d)}$ as it was unneeded and also not correct.